

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 6

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Definition

Let X be a non-negative integer values random variable. The probability generating function Φ_X of X is defined as

$$\Phi_X(t) = \sum_{x=0}^{\infty} P(X = x)t^x, \quad -1 \leq t \leq 1$$

Examples:

- **Binomial Distribution:**

$$\Phi_X = (pt + 1 - p)^n$$

- **Poisson Distribution:**

$$\Phi_X(t) = e^{\lambda(t-1)}$$

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- Let $X_1, \dots, X_d$ are independent, Poisson random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_d$. Then $\sum_{i=1}^n X_i$ has Poisson distribution with parameter $\sum_{i=1}^n \lambda_i$.

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- If X is nonnegative random variable, then the fact that X has finite expectation is usually denoted by $\mathbb{E}[X] < \infty$.

- **Poisson Distribution:** $\mathbb{E}[X] = \lambda$
- **Geometric Distribution:** $\mathbb{E}[X] = \frac{1}{p}$
- **Binomial Distribution:** $\mathbb{E}[X] = np$
- **Bernoulli Distribution:** $\mathbb{E}[X] = p$
- **Distribution not having finite mean:**

$$P(X = x) = \begin{cases} \frac{1}{x(x+1)}, & x = 1, 2, \dots \\ 0, & \text{elsewhere} \end{cases}$$

Expectation of Bernoulli Random Variable

- X is a Bernoulli random variable with following PMF

$$P(X = x) = \begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \end{cases}$$

- $\mathbb{E}[X] = 1 \times P(X = 1) + 0 \times P(X = 0) = 1 \times p + 0 \times (1 - p) = p$

Expectation of Binomial Random Variable

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$$\begin{aligned}\mathbb{E}[X] &= \sum_{k=0}^n k P(X = k) = \sum_{k=0}^n k {}^n C_k p^k (1 - p)^{n-k} \\&= \sum_{k=1}^n k {}^n C_k p^k (1 - p)^{n-k} \\&= \sum_{k=1}^n p k \frac{n}{k} {}^{n-1} C_{k-1} p^{k-1} (1 - p)^{(n-1)-(k-1)} \quad \because {}^n C_k = \frac{n}{k} {}^{n-1} C_{k-1} \\&= np \sum_{k=1}^n {}^{n-1} C_{k-1} p^{k-1} (1 - p)^{n-k} \\&= np \sum_{m=0}^n {}^n C_k p^k (1 - p)^{n-k} = np\end{aligned}$$

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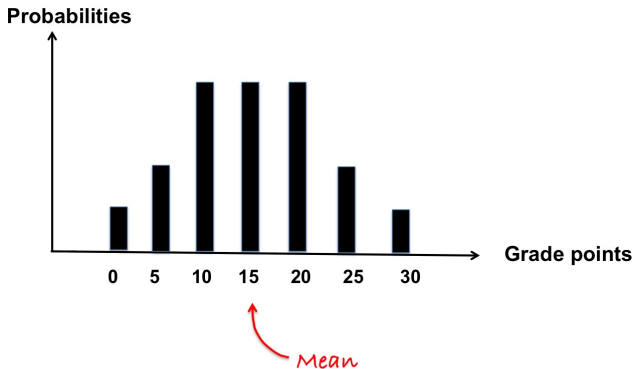
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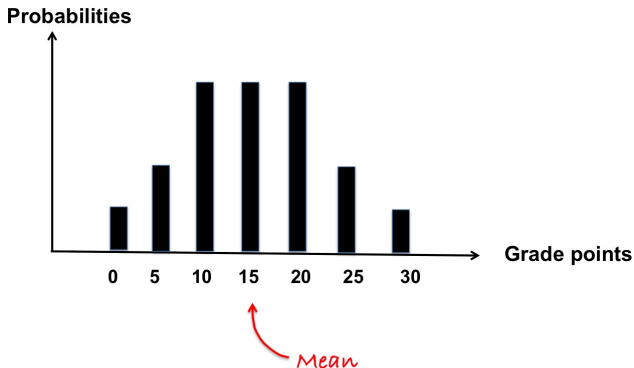
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- $\mathbb{E}[X^k] = \sum_x x^k P(X = x)$. (k^{th} moment of X)

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- How about the spread of the PMF? Is there a way to capture it?

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- The grade points have average 20 out of 30 with a standard deviation of 5 grade points. Roughly this means, most of the students have grade points within $[20 - 5, 20 + 5]$.

Computing the Variance

- $\text{var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \sum_x (x - \mathbb{E}[X])^2 P(X = x)$
- Always remember! $\mathbb{E}[X]$ or $\mathbb{E}[g(X)]$ do not depend on any particular value of x . You can treat it as a constant. It only depends on the PMF of X .
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Proof?

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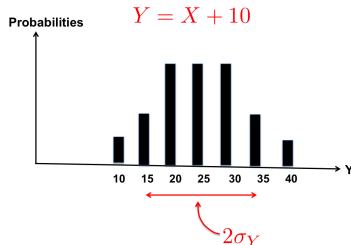
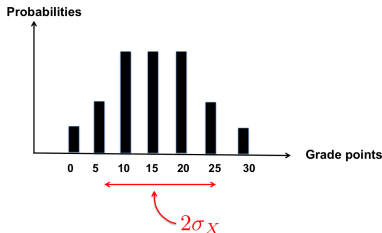
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- $Y = aX^3 + bX^2 + cX + d$. Can you guess what $\mathbb{E}[Y]$ is?
 $\mathbb{E}[Y] = a\mathbb{E}[X^3] + b\mathbb{E}[X^2] + c\mathbb{E}[X] + d$.

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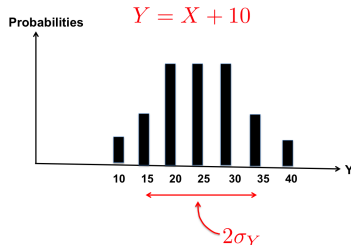
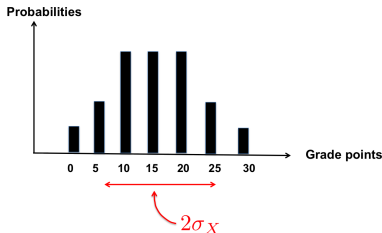
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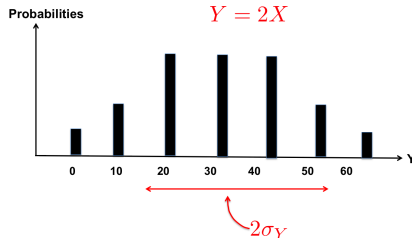
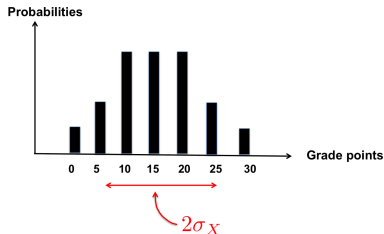
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- So? the spread of the numbers should stay the same!

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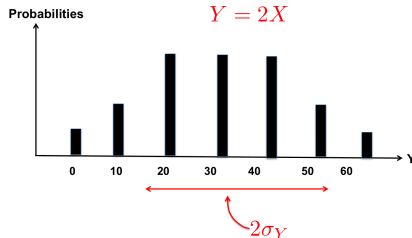
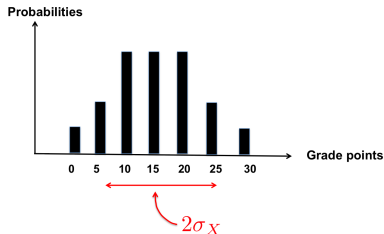
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- So, the spread should increase if $a > 1$!

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- $\text{var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p)$.

- The conditional PMF of a random variable X , conditioned on a particular event A with $P(A) > 0$, is defined by:

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- Total probability rule gives: $P(A) = \sum_x P(\{X = x\} \cap A)$,
 $P(X = x|A) = 1$.

- Conditional expectation of random variable X conditioned on event A is written as $\mathbb{E}[X|A]$

$$\mathbb{E}[X|A] = \sum_x xP(X = x|A)$$